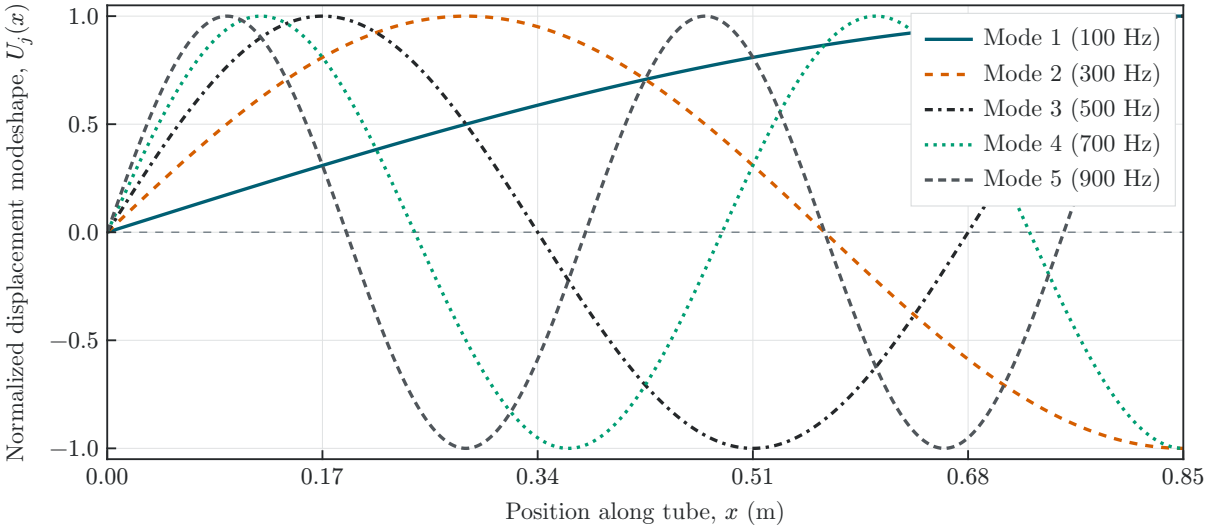

Mode 1 (100 Hz)
Mode 2 (300 Hz)
Mode 3 (500 Hz)
Mode 4 (700 Hz)
Mode 5 (900 Hz)
Normalized displacement modeshape, $U_j(x)$
1.0
0.5
0.0
−0.5
−1.0
0.00
0.17
0.34
0.51
0.68
0.85
Position along tube, x (m)